Instrument selection followed STROBE-MR; counts are tallied from the deposited PLINK clump files and harmonised instrument tables.

Step 1 · Exposure GWAS summary statistics
5 regional BMD sites (Qian 2025; N=31,873 per site)
+ radiographic knee OA (GO Consortium 2025; N≈1.21M)

↓

Step 2 · Genome-wide significant SNPs ($p < 5 \times 10^{-8}$)
170–514 per BMD site; 1,980 knee OA

↓

Step 3 · Independent instruments after strict LD clumping
r^2<0.001 / 10 Mb (1000G EUR; PLINK-benchmarked)
20–35 per BMD site; 116 knee OA

↓

Step 4 · Harmonised with each outcome GWAS
drop palindromic / strand-ambiguous / absent-from-outcome
16–31 (BMD→knee OA); 31, 25 (BMD→fracture);
69 (knee OA as exposure; −41% from 116)

↓

Step 5 · 10 MR pairs analysed
IVW fixed/random, MR-Egger, weighted median,
MR-PRESSO, leave-one-out; MVMR + height on 7 pairs

Instrument attrition by analysed pair

Exposure → outcome	GWS	Clumped	K	Retained
BMD total → knee OA	514	35	**31**	89%
BMD total → fracture	514	35	**31**	89%
Femoral neck → knee OA	170	28	**25**	89%
Femoral neck → fracture	170	28	**25**	89%
Lumbar spine → knee OA	301	24	**21**	88%
Arms → knee OA	246	20	**16**	80%
Legs → knee OA	277	26	**24**	92%
Knee OA → BMD total	1,980	116	**69**	59%
Knee OA → femoral neck	1,980	116	**69**	59%
Knee OA → fracture	1,980	116	**69**	59%

GWS = genome-wide significant SNPs ($p < 5 \times 10^{-8}$); Clumped = independent instruments after strict LD clumping; K = instruments entering MR after harmonisation; Retained = K / clumped.
Knee OA as exposure loses 41% of instruments at harmonisation (116 → 69); the attrition is reported in Methods and examined in the sensitivity analyses.